

# Your first statgen run on an AoU VM

**20-minute lab**

September 2026

**Goal.** Inspect a genotype dataset, run QC, fit an adjusted association model, interpret its output, and save the run. All supplied data are synthetic. Work in the terminal of your running Workbench JupyterLab app.

## 0–3 minutes: orientation and preflight

Open a Terminal from the JupyterLab Launcher. If the instructor has not already staged the folder, clone it (the trailing name keeps the folder called `aou_cli_workshop`):

```
git clone https://github.com/jeffreyswigert/aou-statgen-workshop.git aou_cli_workshop
pwd
ls -lh
cd aou_cli_workshop
bash scripts/00_preflight.sh
```

If already inside the workshop folder, omit `cd`. If PLINK is in the package's tools folder, use `export PLINK2="$PWD/tools/plink2"`. Preflight should find 600 unique, matched synthetic samples. Stop and ask for help if it fails.

```
head -n 5 data/toy.pvar
head -n 5 data/phenotypes.tsv
less scripts/02_qc.sh
```

Press `q` to exit `less`. The `.pgen` file is binary; inspect its companion metadata rather than printing its bytes. In this synthetic dataset the sample key is IID. Real data may require a FID/IID pair.

## 3–8 minutes: QC

```
bash scripts/02_qc.sh
cat results/qc_counts.tsv
head -n 5 results/raw.smiss
head -n 5 results/raw.vmiss
```

The script measures baseline frequencies and missingness, keeps eligible autosomal biallelic SNPs, removes samples with more than 5% missing calls, then filters variants at 2% missingness, MAF 5%, and MAC 20. These are teaching choices, not a complete AoU QC protocol.

**Write your checkpoint:** retained samples \_\_\_\_\_; retained variants \_\_\_\_\_.

Which report describes missing calls for each person? \_\_\_\_\_

**8–14 minutes: association**

```
bash scripts/03_association.sh
cat results/top_hits.tsv
cat results/analysis_summary.txt
less results/assoc.log
```

The model uses the quantitative trait with age, a simulated binary covariate, and five supplied PCs. Each variant enters separately. The trait remains in its original units; the script rescales selected covariates for numerical stability. `omit-ref` fixes the tested allele to ALT for these biallelic sites.

| Column  | Meaning                                                                 |
|---------|-------------------------------------------------------------------------|
| A1      | Allele counted in the effect estimate                                   |
| OBS_CT  | Number of observations used at that variant                             |
| BETA    | Trait-unit difference per additional A1 copy, conditional on covariates |
| SE      | Standard error of the estimated effect                                  |
| P       | P-value for the tested effect                                           |
| ERRCODE | A dot means no reported test error                                      |

**14–18 minutes: explain your result**

1. Name the leading variant and counted allele.
2. State the effect in a complete sentence, including its units and adjustment.
3. Why might OBS\_CT be smaller than the number of QC-passing samples?
4. Compare  $P = 0.001$  with  $0.05/2330$ . Does it pass this correction?
5. Why does including PCs not resolve every concern about relatedness?

**18–20 minutes: save and finish**

```
bash scripts/04_save.sh
```

This requires `WORKSHOP_BUCKET` in `config.sh` to be your authorized workspace’s real `gs://` bucket. The script first runs `scripts/check_disclosure.py`, which screens `results/` for participant-level files and for counts of 1–20 that the AoU dissemination policy restricts, including counts *derivable by subtraction* between reported numbers. Read its two findings on today’s synthetic tables and explain what each one would reveal. With real data (`DATA_MODE=aou`) a finding blocks the save until resolved or explicitly acknowledged. The script then saves a uniquely named archive, verifies its cloud listing, and includes code, logs, the disclosure report, input checksums, and provenance. Retain the original input files too.

After verifying the save, stop the cloud app in the Workbench. If the save fails, preserve the VM files while the instructor resolves access. Closing the browser is not the same as stopping the app.

**Real AoU data:** keep participant-level records and sensitive results inside the authorized Workbench. Never upload them to GitHub or external chat. Review AoU dissemination requirements before sharing outputs.

## Reference answers: synthetic dataset only

| Stage           | Samples | Variants |
|-----------------|---------|----------|
| Input           | 600     | 2,400    |
| Eligible sites  | 600     | 2,390    |
| Sample filter   | 588     | 2,390    |
| Variant filters | 588     | 2,330    |

Ten sites have a simulated LowQual flag. Twelve samples have high missingness. Forty variants have high missingness, and twenty are very rare. The final file set is **results/qc.\***.

The leading association is SIMV001201, with A1 = G, BETA  $\approx$  0.9282, SE  $\approx$  0.06260, N = 587, and P  $\approx$   $2.19 \times 10^{-42}$ . One more G copy is associated with 0.928 trait units conditional on the included covariates. The planted effect was 0.9. The estimate differs because this is one random sample.

One retained individual has a missing call at that variant. This accounts for N = 587 rather than 588. All 2,330 additive tests have valid p-values in the supplied baseline. The simple correction is  $0.05/2330 \approx 2.15 \times 10^{-5}$ , so 0.001 does not pass.

PCs summarize major genotype variation. Family relationships can still induce residual correlation. A real study needs a justified unrelated subset or a model that handles relatives, appropriate phenotype construction, release-specific QC, and a defensible covariate strategy.

## Quick command reference

|                                  |                                                  |
|----------------------------------|--------------------------------------------------|
| <code>pwd / ls -lh</code>        | Where am I? What files are here?                 |
| <code>head -n 5 FILE</code>      | Preview a text file                              |
| <code>less FILE</code>           | Read without printing everything; <b>q</b> exits |
| <code>-pfile PREFIX</code>       | Read a PLINK 2 file set                          |
| <code>-keep / -remove</code>     | Select / exclude sample IDs                      |
| <code>-extract / -exclude</code> | Select / exclude variant IDs                     |
| <code>-mind / -geno</code>       | Sample / variant missingness thresholds          |
| <code>.smiss / .vmiss</code>     | Sample / variant missingness reports             |
| <code>.log</code>                | Inputs, options, counts, warnings, and errors    |

## Stretch exercise: a real phenotype (Controlled Tier)

Finished early? Pick a phenotype from the menu and summarize it from the live CDR:

```
bash scripts/06_pheno_summary.sh          # show the menu
bash scripts/06_pheno_summary.sh height   # or ldl, education, ...
```

Requires CDR\_DATASET and BILLING\_PROJECT in `config.sh` (your instructor supplies both). The script queries **aggregates only**, caps the bytes each query may scan, and suppresses counts of 1–20 on output. Three things to do with the result: (1) check the *resolved concept names* at the top — a concept ID is a claim until the name confirms it; (2) read the trap note for your phenotype’s source type (program measurement vs. EHR lab vs. survey vs. EHR condition — each fails differently); (3) find your phenotype in the public Data Browser ([databrowser.researchallofus.org](http://databrowser.researchallofus.org)) and compare its participant count to yours. To add one as a covariate in a real analysis, this concept-ID-plus-aggregate check is the first step;

a real phenotype still needs units, dates, and a one-row-per-person rule. These are real participant data: summaries stay in the workspace unless they clear dissemination review.

### Sources and next steps

Read the supplied Mills, Barban, and Tropf (2020) Chapters 7–8 alongside the [PLINK 2 documentation](#). Consult the [AoU Data Dictionaries](#) for the selected release. Instructor preparation for real data is in `docs/AOU_ADAPTATION.md`; all source links are in `docs/SOURCES.md`.